

# HW1: A Quarto tutorial + intro to R

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## Practice Exercises with Palmer Penguins

### Setup (5 points)

Load the required package and examine the data:

```
library(palmerpenguins)
```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

penguins, penguins\_raw

```
# Your code here: Explore the dataset structure
# How many rows? How many columns? What variable types?
```

```
penguins <- palmerpenguins::penguins
print(paste("It has", dim(penguins)[1], "rows."))
```

```
[1] "It has 344 rows."
```

```
print(paste("It has", dim(penguins)[2], "columns."))
```

```
[1] "It has 8 columns."
```

```
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm  : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

## Part 1: Vectors and Summary Statistics (25 points)

### 1.1 Extract and analyze bill lengths

```
# Extract bill_length_mm as a vector
#bill_lengths <- # Your code here
bill_lengths <- penguins$bill_length_mm

# Calculate: mean, median, sd, min, max (handle NAs!)
print(paste("The mean is", mean(bill_lengths, na.rm = TRUE)))
```

```
[1] "The mean is 43.9219298245614"
```

```
print(paste("The median is", median(bill_lengths, na.rm = TRUE)))
```

```
[1] "The median is 44.45"
```

```
print(paste("The sd is", sd(bill_lengths, na.rm = TRUE)))
```

```
[1] "The sd is 5.45958371392653"
```

```
print(paste("The min is", min(bill_lengths, na.rm = TRUE)))
```

```
[1] "The min is 32.1"
```

```
print(paste("The max is", max(bill_lengths, na.rm = TRUE)))
```

```
[1] "The max is 59.6"
```

## 1.2 Logical subsetting

```
# Create logical vector: TRUE when bill length > 45mm
#long_bills <- # Your code here
long_bills <- bill_lengths > 45

# How many penguins have long bills?
print(paste(sum(long_bills, na.rm = TRUE), "penguins have long bills"))
```

```
[1] "165 penguins have long bills"
```

## Part 2: Subsetting Data (25 points)

### 2.1 Island analysis

```
# Extract bill lengths for Dream island penguins only
# Calculate the mean for this subset
dream_bill_length <- penguins[penguins$island == "Dream",]$bill_length_mm
print(paste("The mean of the subset is", mean(dream_bill_length, na.rm = TRUE)))
```

```
[1] "The mean of the subset is 44.1677419354839"
```

### 2.2 Species counting

```
# How many Adelie penguins are in the dataset?
print(paste(nrow(penguins[penguins$species == "Adelie",]),
            "Adelie penguins are in the dataset"))
```

```
[1] "152 Adelie penguins are in the dataset"
```

## 2.3 Find the heaviest penguin

```
# Find: body mass, species, and island of the heaviest penguin
penguins[penguins$body_mass_g == max(penguins$body_mass_g, na.rm = TRUE) &
  !is.na(penguins$body_mass_g),
  c("body_mass_g", "species", "island")]
```

```
# A tibble: 1 x 3
  body_mass_g species island
    <int> <fct>   <fct>
1      6300 Gentoo  Biscoe
```

## Part 3: Matrices and apply() (25 points)

### 3.1 Create measurement matrix

```
# Remove rows with missing values
#penguins_complete <- penguins[complete.cases(penguins), ]
penguins_complete <- penguins[complete.cases(penguins), ]

# Create matrix with: bill_length_mm, bill_depth_mm,
# flipper_length_mm, body_mass_g
#penguin_measurements <- # Your code here
penguin_measurements <- as.matrix(penguins_complete[,c("bill_length_mm",
  "bill_depth_mm", "flipper_length_mm", "body_mass_g")])
```

### 3.2 Apply functions

```
# Use apply() to calculate mean of each column
apply(penguin_measurements, 2, mean)
```

| bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g |
|----------------|---------------|-------------------|-------------|
| 43.99279       | 17.16486      | 200.96697         | 4207.05706  |

```
# Use apply() to find which row has the maximum for each column
apply(penguin_measurements, 2, which.max)
```

| bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g |
|----------------|---------------|-------------------|-------------|
| 179            | 15            | 209               | 164         |

## Part 4: Custom Function (20 points)

### 4.1 Write standardize function

```
# Function that: removes NAs, returns (x - mean) / sd
#standardize <- function(x) {
  # Your code here
#}

standardize <- function(x) {
  x_complete <- x[!is.na(x)]
  return ((x_complete - mean(x_complete))/sd(x_complete))
}

# Test on bill_length_mm
#bill_length_standardized <- standardize#(penguins$bill_length_mm)
bill_length_standardized <- standardize(penguins$bill_length_mm)

# Verify: mean 0, sd 1
print(paste("The standardized mean is", mean(bill_length_standardized)))
```

```
[1] "The standardized mean is -6.88208424621442e-17"
```

```
print(paste("The standardized sd is", sd(bill_length_standardized)))
```

```
[1] "The standardized sd is 1"
```

### Bonus (10 points max)

Choose ONE bonus challenge:

**Option A:** Create a summary data frame with one row per species showing: count, mean body mass, mean flipper length

**Option B:** Calculate bill length-to-depth ratio for each penguin. Which species has the highest average ratio?

```
# Your bonus code here
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
# Option A
```

```
penguins |>
  select(species, body_mass_g, flipper_length_mm) |>
  group_by(species) |>
  summarise(count = n(),
            mean_body_mass = mean(body_mass_g, na.rm = TRUE),
            mean_flipper_length = mean(flipper_length_mm, na.rm = TRUE)
  )
```

```
# A tibble: 3 x 4
```

|   | species   | count | mean_body_mass | mean_flipper_length |
|---|-----------|-------|----------------|---------------------|
|   | <fct>     | <int> | <dbl>          | <dbl>               |
| 1 | Adelie    | 152   | 3701.          | 190.                |
| 2 | Chinstrap | 68    | 3733.          | 196.                |
| 3 | Gentoo    | 124   | 5076.          | 217.                |

```
# Option B
```

```
penguins <- penguins |>
  mutate(l2d = bill_length_mm / bill_depth_mm)
```

```
penguins |>
  select(species, l2d) |>
  group_by(species) |>
  summarise(mean_l2d = mean(l2d, na.rm = TRUE)) |>
  arrange(desc(mean_l2d))
```

```
# A tibble: 3 x 2
  species mean_l2d
  <fct>      <dbl>
1 Gentoo    3.18
2 Chinstrap 2.65
3 Adelie    2.12
```

Gentoo has the highest average bill length-to-depth ratio.